Mock Paper A(2)

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Time taken	44 secs
Marks	0.00/109.00
Grade	**0.00** out of 100.00
Feedback	You are on the right track. Could you spend some more time on your revision?

Question 1
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 003
What is the incidence of seizures in patients taking clozapine at a dose of more than 600mg/day?

Select one:
- 0.15
- 0.05
- 0.02
- 0.01
- 0.005

Your answer is incorrect.
Risk of seizures at a dose less than 300 mg/d: 1% to 2%. Dose 300 to 600 mg/d: 3% to 4%. Dose more than 600 mg/d: 5%.
The correct answer is: 0.05

Question 2
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 007
Neuroleptic malignant syndrome (NMS) is associated with a mortality rate of

Select one:
- 15-20%.
- 5-10%
- 25-50%
- 1-2%
- 2-5%

Your answer is incorrect.
5-20% of patients on oral medication and up to 30% of patients on depot formulations who develop the syndrome will die from the condition if untreated.
The correct answer is: 15-20%.

Question 3
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 015
Which one among the following is not a recognised side effect of Valproate?

Select one:
- Agranulocytosis
- Hair loss with curly regrowth
- Tremor
- Weight gain
- Thrombocytopenia

Your answer is incorrect.
Common side effects of sodium valproate include weight gain, nausea, vomiting, diarrhoea, sedation, tremor and hair loss with curly regrowth.
The correct answer is: Agranulocytosis

Question 4
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 016
Mr.Z is a 25 year old gentleman being treated for first episode schizophrenia. He was initially started on Risperidone 2mgs daily which was subsequently increased to 4mgs by week 4. Although he was noted to show good improvement with his delusions and hallucinations, the nurses on the ward noticed he is generally restless, constantly pacing up and down, foot stamping when seated, rocking back and forth and fidgeting in his chair. The nurses wondered whether this could be a side effect of his medication. Which of the following side effects of treatment is he experiencing?

Select one:
- Neuroleptic malignant syndrome
- Akathisia
- Tardive dyskinesia
- Restless leg syndrome

- Pseudo Parkinsonism

Your answer is incorrect.

Akathisia is subjectively an unpleasant state of motor restlessness, which is seen in 20-25% of patients prescribed with anti-psychotic medication. It is a subjectively unpleasant state of inner restlessness where there is a strong desire or compulsion to move. It may become prominent as the dose of antipsychotic medication is increased. Akathisia can be mistaken for psychotic agitation and has been linked with suicide and aggression towards others.

The correct answer is: Akathisia

Question 5

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 019
A significant prolactin elevating effect is seen with one of the following

Select one:

- Ziprasidone
- Clozapine
- Quetiapine
- Aripiprazole
- Amisulpiride

Your answer is incorrect.

Maudsley Prescribing Guidelines. 9th ed. p 10.

The correct answer is: Amisulpiride

Question 6

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 020
The antidepressant most associated with QTc prolongation is

Select one:

- Venlafaxine
- Paroxetine
- Mirtazapine
- Amitriptyline
- Moclobemide

Your answer is incorrect.

The antidepressant class most associated with QTc prolongation is Tricyclic antidepressants such as Amitriptyline. Moclobemide, citalopram, and venlafaxine have a low effect on QTc prolongation. Mirtazapine and SSRIs (except citalopram) have no effect on QTc prolongation

The correct answer is: Amitriptyline

Question 7

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 021
Which of the following psychotropics has a tendency to reduce gastrointestinal transit with a resultant slow absorption when given orally?

Select one:

- Amitriptyline
- Risperidone
- Diazepam
- Fluoxetine
- Lithium carbonate

Your answer is incorrect.

Amitriptyline, due to its atropine-like anticholinergic action, can slow GI transit with a resultant slow absorption. However, nearly 95% of the orally administered drug gets absorbed.

The correct answer is: Amitriptyline

Question 8

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 022
Mrs. Y is a 28-year-old pregnant woman with severe depression. The ideal anti-depressant treatment of choice would be

Select one:

- Venlafaxine
- Reboxetine
- Paroxetine
- Fluvaxamine
- Amitryptyline

Your answer is incorrect.

The antidepressants recommended for treatment of depression in pregnancy are Amitriptyline, nortryptyline, imipramine and Fluoxetine

The correct answer is: Amitryptyline

Question 9
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 023
A 65-year-old man develops recurrent UTI since starting an antidepressant. He also gives a history suggestive of new onset prostatism. But prostate examination shows no enlargement. The most likely offending agent is

Select one:
- Nefazadone
- Reboxetine
- Trazodone
- Mirtazapine
- Amitryptyline

Your answer is incorrect.

Drugs with significant anticholinergic properties such as amitriptyline are notorious to cause signs of prostatism in elderly males. They may produce urinary retention and sometimes bladder distension.

The correct answer is: Amitryptyline

Question 10
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 024
Tom is an inpatient on a psychiatric ward. He has a history of psychotic depression and is on combination treatment with antipsychotics and antidepressants. He appears slightly flushed, with dry mouth, dry skin, feeling hot, and confused. He is exhibiting features of

Select one:
- SSRI discontinuation syndrome
- Neuroleptic malignant syndrome
- Idiopathic parkinsonism
- Anticholinergic syndrome
- Serotonin syndrome

Your answer is incorrect.

Anticholinergic syndrome results from the inhibition of cholinergic neurotransmission at muscarinic receptor sites. Psychotropics such as phenothiazines, tricyclic antidepressants and antiparkinsonian agents have anticholinergic properties. When they are administered in combination, the action of these drugs becomes additive resulting in anticholinergic syndrome. Features of a mild organic brain syndrome, which includes disorientation, difficulty in concentrating and impaired short-term memory are seen, especially noticeable at nighttime. Other physical symptoms include flushing, dry skin and mucosa, mydriasis with loss of accommodation and fever (Ref: Kaplan & Sadock Synopsis of Psychiatry 9th ed. Pg 1170-1172)

The correct answer is: Anticholinergic syndrome

Question 11
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 025
Which drug increases the risk of bleeding in patients on SSRIs?

Select one:
- Paracetamol
- Aspirin
- Naproxen
- Ibuprofen
- Diclofenac

Your answer is incorrect.

There is an increased risk of bleeding when SSRIs are taken with antiplatelet drugs such as low-dose aspirin or clopidogrel and non-steroidal anti-inflammatory drugs (NSAIDs). However, the risk is particularly high with aspirin than other NSAIDs. A large Canadian study found patients taking both aspirin and an SSRI had a 42% increased risk of a bleeding episode - gastrointestinal bleeding, hemorrhagic stroke or other forms of bleeding that required hospital admission - compared with those taking aspirin alone. (Labos et al., 2011: http://www.cmaj.ca/content/early/2011/09/26/cmaj.100912)

The correct answer is: Aspirin

Question 12
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 026
A 33 year old man is admitted to Psychiatric Intensive Care Unit with an acute relapse of schizophrenia. He is started on a typical antipsychotic medication after which his symptoms reduce in intensity. But since this new treatment was started, he complains of feeling restless, and is seen pacing up and down a lot. When you interview him, he appears to be shuffling often and not able to stay for long in the interview room. If you want to prescribe propranolol to this patient, which of the following medical condition will be a contraindication for the prescription?

Select one:
- Asthma
- Obesity
- Hypertension
- Akathisia
- Elevated eosinophils

Your answer is incorrect.

Propranolol and other beta-blockers are recommended to treat akathisia, but beta-receptor blockade causes bronchospasm, and so are contraindicated in asthma.

The correct answer is: Asthma

Question 13
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 027
Which of the following is not a feature of Neuroleptic Malignant Syndrome?

Select one:
- Incontinence
- Bradycardia
- Tremor
- Rigidity
- Diaphoresis

Your answer is incorrect.

Bradycardia is not a feature of NMS. Neuroleptic Malignant Syndrome can occur at anytime during treatment with neuroleptics. It consists of the tetrad of extreme hyperthermia, severe muscular rigidity and confusion, and autonomic fluctuations (BP and pulse rate). Other clinical features of neuroleptic malignant syndrome include the following diaphoresis, dysphagia, tremor, shuffling gait, tachycardia, dyspnoea, incontinence, delirium progressing to lethargy, stupor, coma

The correct answer is: Bradycardia

Question 14
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 028
Which of the following is the most likely change seen in the ECG of patients taking lithium?

Select one:
- Flattened trace
- ST elevation
- Prolonged QRS
- Prolonged QTc
- Changes in T wave morphology

Your answer is incorrect.

The most frequently observed change during treatment was flattening or inversion of the T wave. Heart rate decreased, and P-Q interval increased slightly during therapy. There was no change in the S-T segment or the Q-T interval corrected for heart rate. Well monitored lithium treatment of patients with serum lithium concentrations of about 0.6 mmol/l does not cause serious ECG abnormalities. In the absence of symptoms or signs of heart disease, routine monitoring of ECG is not necessary during lithium treatment. (Ref: Bucht G, Smigan L, Wahlin A, Eriksson P. ECG changes during lithium therapy. A prospective study. Retrieved from http://onlinelibrary.wiley.com/doi/10.1111/j.0954-6820.1984.tb03777.x/)

The correct answer is: Changes in T wave morphology

Question 15
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 029
Patients on tranylcypromine should not take an excess of

Select one:
- Swede
- Banana
- Cheese
- Potatoes
- Salt

Your answer is incorrect.

Cheese can induce tyramine related reaction in those on MAOIs.

The correct answer is: Cheese

Question 16
Not answered
Marked out of 1.00
Flag question

MoK Basic Pharmacology 006
Which among the following antipsychotic drugs causes prolongation of QT-c interval on the ECG?

Select one:
- Clozapine
- Olanzapine
- Haloperidol
- Chlorpromazine
- Thioridazine

Your answer is incorrect.

The effect on QTC prolongation is high for some psychotropic drugs like Thioridazine, Pimozide, Sertindole, and any intravenous antipsychotic. There is a moderate effect on QT-c prolongation for psychotropic drugs such as chlorpromazine, quetiapine, ziprasidone and tricyclic antidepressants.

The correct answer is: Thioridazine

Question 17
Not answered
Marked out of 1.00
Flag question

MoK Pharmacodynamics 006
Which of the following is not true regarding drugs used to prevent relapse in alcohol dependence?

Select one:
- Acamprosate interacts with Disulfiram

- Disulfiram inhibits metabolism of warfarin
- Metronidazole can cause disulfiram like reactions with alcohol
- Alcohol should be avoided for 1week following cessation of disulfiram
- Acamprosate acts on NMDA receptors

Your answer is incorrect.

Acamprosate does not interact with Disulfiram. It is believed to act through enhancing GABA transmission in the brain. It is licensed to prevent relapse to alcohol use and has been found to have a modest treatment effect.

The correct answer is: Acamprosate interacts with Disulfiram

Question 18

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 007
Choose one drug that modulates melatonin system directly.

Select one:

- Trazodone
- Zopiclone
- Agomelatine
- Perphenazine
- Galantamine

Your answer is incorrect.

Agomelatine is a new antidepressant with a novel mechanism of action. It acts as an agonist to both the melatonin receptors MT1 and MT2. It is also an antagonist to the 5HT2C serotonin receptor.

The correct answer is: Agomelatine

Question 19

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 008
Which of the following accurately reflects the activity of benzodiazepines?

Select one:

- Reduce GABA transmission
- Block sodium reuptake
- Allosteric modification of GABA receptor
- Substitute for GABA
- Facilitate action potential

Your answer is incorrect.

Benzodiazepines (BDZs) bind to the gamma sub-unit of the GABA-A receptor. Their binding causes an allosteric (structural) modification of the receptor that results in an increase in GABA-A receptor activity. BDZs do not substitute for GABA, which bind at the alpha sub-unit, but increase the frequency of channel opening events which leads to an increase in chloride ion conductance and inhibition of the action potential. (Ref: From CNSforum: Basic and clinical pharmacology, 8th edition. Katzung BG. USA: The McGraw Hill Companies, Inc, 2001:364-381.)

The correct answer is: Allosteric modification of GABA receptor

Question 20

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 009
A 62 year old patient with insomnia and depression develops priapism when trazodone is prescribed. This is possibly mediated via

Select one:

- Beta 1 agonism
- Alpha 2 antagonism
- Alpha 1 antagonism
- Alpha 2 agonism
- Alpha 1 agonism

Your answer is incorrect.

Alpha 1 blockade causes priapism when using trazodone or risperidone.

The correct answer is: Alpha 1 antagonism

Question 21

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 010
Impotence as a side effect of antipsychotics is caused due to which of the following mechanisms?

Select one:

- Alpha-2 adrenergic blockade
- Histaminergic receptor blockade
- Nicotinic receptor blockade
- Alpha-1 adrenergic blockade
- Muscarinic receptor blockade

Your answer is incorrect.

Other side effects caused due to alpha 1 adrenergic blockade includes sedation, postural hypotension, and ejaculatory failure.Alpha 1 adrenergic blockade leads to a moderately high reported incidence of ejaculatory problems such as retrograde ejaculation and rarely priapism. Sexual dysfunction has been reported as a side effect of all antipsychotics and up to 45% of people taking conventional antipsychotics experience sexual dysfunction.

The correct answer is: Alpha-1 adrenergic blockade

Question 22

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 011
The mechanism of action of Varenicline is

Select one:

- Alpha-2, beta-4, nicotinic acetylcholine receptor partial agonist
- Alpha-2, beta-2, nicotinic acetylcholine receptor partial agonist
- Alpha-4, beta-2, nicotinic acetylcholine receptor partial antagonist
- Alpha-4, beta-2, nicotinic acetylcholine receptor partial agonist
- Alpha-2, beta-4, nicotinic acetylcholine receptor partial antagonist

Your answer is incorrect.

Varenicline is a partial agonist at the alpha-4 beta-2 unit of nicotinic acetylcholine receptor. It assists smoking cessation by relieving nicotine withdrawal symptoms and reducing the rewarding properties of nicotine

The correct answer is: Alpha-4, beta-2, nicotinic acetylcholine receptor partial agonist

Question 23

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 012
Which of the following medications have dual action in prophylaxis of influenza as well as in treatment of Parkinson's disease?

Select one:

- Carbidopa
- Levodopa
- Orphenadrine
- Bromocriptine
- Amantadine

Your answer is incorrect.

Amantadine is an antiviral agent and prevents viral uncoating. It also induces dopamine release from nerve terminals and so is used in Parkinsonism.

The correct answer is: Amantadine

Question 24

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 007
What is the half-life of risperidone?

Select one:

- 2 hours
- 5 hours
- 150 hours
- 8 hours
- 24 hours

Your answer is incorrect.

The half-life of risperidone and of its active metabolite is about 24 hours.

The correct answer is: 24 hours

Question 25

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 008
Mr. Z is a 43 year old gentleman who has a 15 year history of schizophrenia and poor adherence with antipsychotic treatments. He has been stabilised on 15mgs of Olanzapine in the hospital and he has previously done well on Flupentixol injections. He has agreed to switch to Flupentixol injection once a fortnight. The test dose was administered. What is the most likely amount of time he will need to continue taking the oral Olanzapine?

Select one:

- 3- 6 months
- 2- 3 months
- 2-3 weeks
- 1 month
- 1- 2 weeks

Your answer is incorrect.

The depot medication may require 3-6 months to reach a steady plasma level, and, therefore, oral antipsychotic medication should be continued until steady state plasma level is achieved. Attainment of peak plasma levels, therapeutic effect and steady state plasma levels are all delayed with depot injection. The use of adjunctive oral medication to assess depot requirements may be helpful and oral anti psychotic cover is sometimes required for longer periods of time. Moreover, in most clinical trials relapse seems only to occur six months after withdrawing depot therapy; roughly the time required to clear steady state depot drug levels from the blood.

The correct answer is: 3- 6 months

Question 26
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 009
The dosage of duloxetine used for the treatment of diabetic neuropathy is

Select one:
- 60-120 mg
- 5-10 mg
- 10-20 mg
- 120-240 mg
- 30-60 mg

Your answer is incorrect.

Duloxetine inhibits the re-uptake of both serotonin and noradrenaline. It is used for the treatment of major depressive disorder (60 mg once daily), generalized anxiety disorder (60-120 mg) and the treatment of diabetic neuropathy (60-120 mg in divided doses).

The correct answer is: 60-120 mg

Question 27
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 010
Aripiprazole has an elimination half life of

Select one:
- 3.5 hours
- 2 hours
- 12 hours
- 24 hours
- 75 hours

Your answer is incorrect.

Aripiprazole is metabolized extensively in humans, with a mean terminal half-life of 60 hours (up to 76 hours) after 14 days of administration. The half-life of its major metabolite, dehydroaripiprazole, is 96 hours.

The correct answer is: 75 hours

Question 28
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 011
Which of the following is considered to be the therapeutic blood level of clozapine?

Select one:
- Above 550 ng/mL
- above 450 ng/mL
- above 350 ng/mL
- Above 250 ng/mL
- Above 150 ng/mL

Your answer is incorrect.

Blood levels above 350 ng/mL are believed to be correlated with therapeutic response.

The correct answer is: above 350 ng/mL

Question 29
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 012
Which of the following follows zero order kinetics?

Select one:
- Sodium valproate
- Lorazepam
- Alcohol
- Haloperidol
- Amitriptyline

Your answer is incorrect.

A drug with zero order kinetics is more likely to achieve toxic levels than a drug with first order kinetics because its rate of elimination cannot increase beyond a certain level. E.g. Alcohol.

The correct answer is: Alcohol

Question 30
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 015
Which of the following antipsychotics is primarily renally excreted?

Select one:
- Olanzapine
- Amisulpride
- Clozapine
- Haloperidol

- Risperidone

Your answer is incorrect.

Amisulpride is primarily renally excreted. Nevertheless, there is a limited experience of using any antipsychotics it is in renal disease and, therefore, best to avoid them if possible in well established renal failure.

The correct answer is: Amisulpride

Question 31
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 017
A 23 year old patient is observed to consume high amounts of coffee whilst in the ward. She is being treated for a depressive episode. Which of the following statement is likely to be true in this case?

Select one:
- Caffeine is poorly absorbed orally
- Caffeine increases sodium retention
- Caffeine reduces renal clearance of lithium
- Caffeine can interact with drugs that are metabolised by CYP1A2
- Caffeine suppresses weight gain

Your answer is incorrect.

Controlling caffeine intake when investigating renal function using lithium clearance is important as caffeine has a natriuretic effect and increases lithium clearance as well. It is metabolised by CYP1A2 and can interact with drugs sharing this mechanism Clin Sci (Lond). 2002 Nov;103(5):461-6.

The correct answer is: Caffeine can interact with drugs that are metabolised by CYP1A2

Question 32
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 018
Which of the following drugs when combined with lithium, do NOT cause an increase in lithium levels per se but still cause increased incidence of neurotoxicity?

Select one:
- ACE inhibitors
- Fluoxetine
- Frusemide
- Calcium channel blockers
- NSAIDs

Your answer is incorrect.

See course notes on mood stabilisers, paper 2 pharmacology. Haloperidol and clozapine can produce increased neurotoxic effects with lithium without altering the plasma levels.

The correct answer is: Calcium channel blockers

Question 33
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 003
Which of the following statement is correct with regard to pharmacokinetics?

Select one:
- Chlorpromazine decreases levels of tricyclics
- Most orally administered antipsychotics undergo first pass metabolism
- Depots achieve peak plasma levels on the third day after injection
- First pass metabolism enables easier entry through blood brain barrier
- Extensive first pass metabolism increases bioavailability

Your answer is incorrect.

First pass metabolism will serve to reduce bioavailability. It does not influence passage via BBB. CPZ inhibits TCA metabolism thereby increasing TCA levels.

The correct answer is: Most orally administered antipsychotics undergo first pass metabolism

Question 34
Not answered
Marked out of 1.00
Flag question

MoK Applied Neuroscience 010
Which one of the following is not a frontal lobe function test?

Select one:
- Tower of London
- Verbal fluency
- Luria's motor tests.
- California Verbal Learning Test
- Cognitive estimates

Your answer is incorrect.

Frontal lobe tests include Stroop test, Halstead trial making, cognitive estimates, Tower of London, Wisconsin card sorting test, verbal fluency, Luria's motor tests and multiple errands tests. The California Verbal Learning Test (CVLT) is useful to asses verbal memory. Though it may be affected in patients with frontal lesions, CVLT is not a specific frontal test.

The correct answer is: California Verbal Learning Test

Question 35
Not answered
Marked out of 1.00
Flag question

MoK Applied Neuroscience 011
Aura is a commonly occurring phenomenon seen in which type of seizures?

Select one:
- Grandmal seizures
- Simple partial seizures
- Complex partial seizures
- Petitmal seizures
- Myoclonic seizures

Your answer is incorrect.

Auras are particularly common in temporal lobe complex partial epilepsy, generally last only a few seconds. Auras are typically stereotyped, e.g., autonomic or visceral aura (epigastric sensations) and may consist of depersonalization, derealisation and affective changes (like anxiety, euphoria), perceptual experiences like illusions and hallucinations.

The correct answer is: Complex partial seizures

Question 36
Not answered
Marked out of 1.00
Flag question

MoK Applied Neuroscience 012
In Broca's area lesion, which of the following features is seen?

Select one:
- Naming is preserved
- Comprehension is preserved
- Inflection is preserved
- Repetition is preserved
- Fluency is preserved

Your answer is incorrect.

Patients with Broca's aphasia produce slow, halting speech that is rarely grammatical. Typical Broca's aphasics eliminate inflections such as -ed and words not central to the meaning of the sentence, such as the and and

The correct answer is: Comprehension is preserved

Question 37
Not answered
Marked out of 1.00
Flag question

MoK Applied Neuroscience 013
A 50-year-old woman began complaining of double vision and blurry vision 3 months ago and has since had diminishing interaction with her family, a paucity of thought and expression, and unsteadiness of gait. Her whole body appears to jump in the presence of a loud noise. An MRI scan and routine CSF examination are unremarkable. What is the most appropriate diagnosis?

Select one:
- Creutzfeldt-Jakob disease
- Alzheimer's disease
- Normal-pressure hydrocephalus
- Parkinson's disease
- Transient global amnesia

Your answer is incorrect.

The neurological com- plaints occurring early in the course of Creutzfeldt-Jakob disease are often cerebellar or visual. Patients may complain of ataxia, clumsiness, or dysarthria, as well as diplopia, distorted vision, blurred vision, field defects, changes in color perception, and visual agnosia. Ultimately cortical blindness may occur.

The correct answer is: Creutzfeldt-Jakob disease

Question 38
Not answered
Marked out of 1.00
Flag question

MoK Applied Neuroscience 014
Raclopride is a compound used in imaging brain receptors. Which of the following receptors are blocked by raclopride?

Select one:
- D2 receptors
- D1 autoreceptors
- 5HT2C receptors
- 5HT3 receptors
- 5HT2A receptors

Your answer is incorrect.

Raclopride is a D2 antagonist that can be radiolabelled with the carbon-11 radioisotope for use in positron emission tomography (PET) scanning to assess the distribution of D2 receptors and the degree of endogenous dopamine binding to this receptor.

The correct answer is: D2 receptors

Question 39
Not answered
Marked out of 1.00
Flag question

MoK Applied Neuroscience 015
Performance sub-sets of WAIS-R include all but

Select one:
- Digit span
- Digit symbol

- Picture arrangement
- Picture completion
- Block design

Your answer is incorrect.

Verbal subtests include similarities, arithmetic, digit span, vocabulary, information and comprehension tests. Performance subtests include picture arrangement, block design, picture completion, digit symbol, matrix reasoning (replaces object assembly).

The correct answer is: Digit span

Question 40

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 016
Which one of the following tests measures non-verbal IQ?

Select one:

- Digit span
- Digit Symbol
- Vocabulary
- Arithmetic
- Information

Your answer is incorrect.

WAIS has both verbal and performance tests. Verbal tests = similarities, arithmetic, digit span, vocabulary, information and comprehension. Performance tests = picture arrangement, block design, picture completion, digit symbol, matrix reasoning (replaces object assembly in newer versions)

The correct answer is: Digit Symbol

Question 41

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 017
Patients with Gerstmann's syndrome usually demonstrate which of the following?

Select one:

- Dominant parietal lobe signs
- Aphasia
- Hemineglect
- Cortical blindness
- Dressing apraxia

Your answer is incorrect.

Damage to the inferior parietal lobule of the dominant hemisphere results in Gerstmann's syndrome. It is characterized by finger agnosia, right-left disorientation, dysgraphia and dyscalculia.

The correct answer is: Dominant parietal lobe signs

Question 42

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 018
Epileptic automatisms usually lasts for

Select one:

- Few minutes to hours
- Few days to months
- Few seconds to minutes
- Few hours to days
- Few months to years

Your answer is incorrect.

Epileptic automatism is a state of clouding of consciousness which occurs during or immediately after a seizure. The impairment of awareness varies. The individual retains control of posture and muscle tone but performs simple or complex movements without being aware of what is happening. To the onlooker, the patient appears confused, and there is subsequent amnesia for the episode. Simple stereotyped behaviours (gesturing, grasping, lip-smacking and chewing movements) are often exhibited lasting few seconds to minutes. Very occasionally, automatisms are prolonged (fugue states), or complex actions are carried out. If violent, these are never premeditated, seldom goal-directed, rarely involves use of complex tools/weapons and are especially likely if restraining was attempted

The correct answer is: Few seconds to minutes

Question 43

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 004
If a 34 year old mother suffers from schizophrenia, what is the lifetime morbid risk for her only child for schizophrenia?

Select one:

- 5 to 10%
- 10 to 15%
- 30 to 40%
- 20 to 30%
- Less than 1%

Your answer is incorrect.

Results of the Copenhagen High-Risk Study (207 children of schizophrenic mothers and 104 control children followed since 1962) indicated that 16.2% of the high-risk children versus 1.9% of the control group developed schizophrenia, and another 4.6% developed a related illness (versus 0.9%. 10-15% is the closest answer to this question.

The correct answer is: 10 to 15%

Question 44

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 007
Mrs. X has adopted a child from family of alcohol dependent parents. The risk of her adopted child suffering from alcohol dependence is increased by

Select one:

- 12 times
- 8 times
- 4 times
- 20 times
- 10 times

Your answer is incorrect.

Adoptees have the risk of alcoholism proportional to the biological family (1.6 to 3.6 times greater in adopted away children of alcoholics). Close family members have 4-fold increased risk of alcoholism.

The correct answer is: 4 times

Question 45

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 008
The odds of having panic disorder is increased in first degree relatives of patients with panic disorder compared to general population by a factor of

Select one:

- 18-20 times
- 2-3 times
- 10-12 times
- 4-8 times
- No increase in prevalence

Your answer is incorrect.

Several family studies have documented an increased panic disorder risk among affected individuals' relatives. Among relatives, the prevalence varies between 6 to 17%, which translates to an average of 4-8 times increase in the risk. If the index cases have onset ages below 20, the risk appears to be much more (17-times). (http://www.ncbi.nlm.nih.gov/pmc/articles/PMC3102861/)

The correct answer is: 4-8 times

Question 46

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 009
A 24 year old woman suffers from bipolar affective disorder. The risk of her twin brother developing bipolar affective disorder is

Select one:

- Above 90%
- 40-50%
- 0-5%
- 5-15%
- 65-70%

Your answer is incorrect.

The risk in dizygotic twins is 5-15%. ('Dizygotic' as it is a twin 'brother'. Twin sister could have been either monozygotic or dizygotic)

The correct answer is: 5-15%

Question 47

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 015
Patients with Down's syndrome have higher risk of Alzheimer's disease. Which of the following gene is implicated in this relationship?

Select one:

- Presenilin 1
- Presenilin 2
- Amyloid Precursor Protein
- Tau
- Apo E4

Your answer is incorrect.

A pathological cascade in both Alzheimer's and Down's -related dementia could be triggered by alterations in APP processing. Several genes that might play a role in the development of AD are found in chromosome 21. Among them are the amyloid precursor protein and the overexpression of the APP gene might be related to the overproduction of the major protein observed in the senile plaque, the A-beta (1-42) peptide, which is considered to be one of the important factors leading to the development of the pathology in Down syndrome with dementia (Alzheimer Disease in Down Syndrome - Medscape Reference. Retrieved from http://emedicine.medscape.com/article/1136117-overview)

The correct answer is: Amyloid Precursor Protein

Question 48
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 019
The isoforms A and B of enzyme Monoamine oxidase are encoded on chromosomes

Select one:
- 21 and 22 respectively
- Chromosomes 1 and 14 respectively
- X and 21 respectively
- Both encoded on chromosome X
- Both encoded on chromosome 14

Your answer is incorrect.
Both isoforms of MAO are encoded on chromosome X. http://www.nature.com/nrg/journal/v6/n10/full/nrg1687.html
The correct answer is: Both encoded on chromosome X

Question 49
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 020
A 25-year-old gentleman has moderate level of learning disability with expressive language disorder, social anxiety, impulsivity, marked inattention. Molecular genetic testing may show an expansion of which repeats;

Select one:
- CAG
- CCCG
- CTG
- CGG
- GCC

Your answer is incorrect.
These are behavioural phenotypes seen in Fragile X syndrome. The syndrome is associated with a large sequence of triplet repeats (CGG) at a fragile site on the X chromosome (Xq27.3)
The correct answer is: CGG

Question 50
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 021
The presence of 2 or more cell lines from different zygotes in a single individual is called

Select one:
- Mosaicism
- Chimerism
- Translocation
- Transcription
- Aneuploidy

Your answer is incorrect.
A chimera or chimaera is a single organism (usually an animal) that is made up of various genetically distinct cells derived from different zygotes. In contrast to mosaic organisms who have different cellular lineages with distinct genetic composition, but all cells originating from the same zygote, chimeric organisms are formed from more than one zygote. Chimeras do not occur naturally; they are formed from four parent cells (two fertilized eggs or early embryos) often fused together in laboratory conditions.
The correct answer is: Chimerism

Question 51
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 022
Transcriptome is a

Select one:
- Statistical test
- Type of plasmid
- Collection of expressed m-RNA
- Positional cloning
- Stretch of DNA

Your answer is incorrect.
A transcriptome is different from a genome, the latter being the entire DNA sequence of an organism. A transcriptome represents the very small percentage of the genome - less than 5 percent in humans - that is transcribed into RNA molecules. A gene may produce many different types of mRNA molecules, so a transcriptome is much more complex than the genome that encodes it.
The correct answer is: Collection of expressed m-RNA

Question 52
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 023
A nonsense mutation refers to

Select one:
- A point mutation
- Substitution of stop codons

- Destruction of stop codons
- Deletion of stop codons
- Creation of stop codons

Your answer is incorrect.

A nonsense mutation prematurely introduces a stop codon in the transcribed mRNA resulting in an incomplete and often nonfunctional protein.

The correct answer is: Creation of stop codons

Question 53
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 024
The gene most strongly implicated in ADHD is

Select one:
- Dopamine 2 receptor gene
- Dopamine 3 receptor gene
- Dopamine 4 receptor gene
- Dopamine 1 receptor gene
- Dopamine 5 receptor gene

Your answer is incorrect.

The gene most strongly implicated in ADHD is the D4 dopamine receptor gene DRD4. The distribution of DRD4 mRNA in the brain suggests that it plays a role in cognitive and executive functions. These functions are implicated in the pathophysiology of ADHD (Patterson 1999)[Retrieved from http://apt.rcpsych.org/content/aptrcpsych/16/2/96.full.pdf]

The correct answer is: Dopamine 4 receptor gene

Question 54
Not answered
Marked out of 1.00
Flag question

MoK Neuroanatomy 006
The structure of the brain involved in taste sensation is

Select one:
- Amygdala
- Hippocampus
- Prefrontal cortex
- Entorhinal cortex
- Anterior insula

Your answer is incorrect.

The primary gustatory cortex is responsible for taste perception. It consists of two substructures: the anterior insula on the insular lobe and the frontal operculum on the inferior frontal gyrus of the frontal lobe. In each hemisphere of the mammalian brain, the insular cortex (often called insular lobe) is a portion of the cerebral cortex folded deep within the lateral sulcus between the temporal lobe and the frontal lobe.

The correct answer is: Anterior insula

Question 55
Not answered
Marked out of 1.00
Flag question

MoK Neuroanatomy 008
Which of the following areas of the brain does not lack blood brain barrier?

Select one:
- Median eminence
- Posterior pituitary
- Area postrema
- Subfornical organ
- Anterior pituitary

Your answer is incorrect.

The anterior pituitary does not lack BBB. There are some areas of the brain that do not have a blood- brain barrier. These are so-called circum-ventricular organs e.g. subfornical organ, area postrema (chemo receptor trigger zone), median eminence and posterior pituitary. Here the capillaries are fenestrated like those in peripheral tissues

The correct answer is: Anterior pituitary

Question 56
Not answered
Marked out of 1.00
Flag question

MoK Neuroanatomy 009
The white matter tract connecting Wernicke's are and Broca's area is called

Select one:
- Cingulate gyrus
- Anterior commissure
- Arcuate fasciculus
- Habenular commissure
- Planum temporale

Your answer is incorrect.

In the dominant hemisphere, arcuate fasciculus connects Wernicke's area and Broca's area. The function of the arcuate fasciculus in right hemisphere is unclear. Lesions of dominant arcuate fasciculus can cause conduction aphasia.' http://www.lib.mcg.edu/edu/eshuphysio/program/section8/8ch15/s8c15_14.htm

The correct answer is: Arcuate fasciculus

Question 57
Not answered
Marked out of 1.00
Flag question

MoK Neuroanatomy 010
Which hypothalamic nucleus regulates prolactin secretion?

Select one:
- Medial dorsal nucleus
- Suprachiasmatic nucleus
- Preoptic nucleus
- Subthalamic nucleus
- Arcuate nucleus

Your answer is incorrect.

The prolactin-releasing factor is found in the arcuate n ucleus of the hypothalamus. This activates the lactotrophic cells of the anterior pituitary. Dopamine, via the tuberoinfundibular dopaminergic system, regulates prolactin secretion through direct projection to the pituitary. Prolactin-secreting microadenomas can be treated by the drug bromocriptine, a dopamine agonist (often used in the treatment of Parkinson's). By giving a dopamine agonist, which acts as Prolactin Inhibiting Factor, the tumour cells are inhibited, and eventually the tumour shrinks.

The correct answer is: Arcuate nucleus

Question 58
Not answered
Marked out of 1.00
Flag question

MoK Neuroanatomy 011
Which of the following cells help in regulating potassium ion concentration in the extracellular space around neuronal cell body?

Select one:
- Purkinje cells
- Astrocytes
- Microglia
- Schwann cells
- Oligodendrocytes

Your answer is incorrect.

Astrocytes are the most numerous type of glial cell. They insulate and separate neurons from one another, and regulate the potassium ion concentration in the extracellular space.'

The correct answer is: Astrocytes

Question 59
Not answered
Marked out of 1.00
Flag question

MoK Neurochemistry 005
Acetylcholine is a substrate of which of the following substances?

Select one:
- Choline Acetyl transferase
- Acetyl cholinesterase
- Monoamine oxidase
- Acetyl Co A
- Choline

Your answer is incorrect.

Acetylcholine is produced by the combination of acetyl coA with choline in the presence of ChAT - choline acetyl transferase. Ach is a substrate of the acetylcholinesterase enzyme.

The correct answer is: Acetyl cholinesterase

Question 60
Not answered
Marked out of 1.00
Flag question

MoK Neurochemistry 006
Which enzyme is required for the metabolism of the transmitter at the neuromuscular junction?

Select one:
- Choline acetyltransferase
- Alpha ketoglutarate
- Glutamine synthetase
- Acetylcholinesterase
- Glutaminase

Your answer is incorrect.

The enzyme involved is degrading acetylcholine is acetylcholinesterase it breaks down ACh into acetate and choline. Choline is then taken up by the presynaptic terminal.

The correct answer is: Acetylcholinesterase

Question 61
Not answered
Marked out of 1.00

MoK Neurochemistry 007
The rate limiting enzyme in alcohol metabolism is

Select one:

Flag question

- Acetaldehyde dehydrogenase
- Alanine dehydrogenase
- DOPA dehydrogenase
- Alcohol dehydrogenase
- Aldehyde dehydrogenase

Your answer is incorrect.

The first step in the metabolism of alcohol is the oxidation of ethanol to acetaldehyde catalyzed by alcohol/dehydrogenase containing the coenzyme NAD+. The acetaldehyde is further oxidized to acetic acid and finally CO_2 and water through the citric acid cycle. A number of metabolic effects from alcohol are directly linked to the production of an excess of both NADH and acetaldehyde

The correct answer is: Alcohol dehydrogenase

Question 62

Not answered

Marked out of 1.00

Flag question

MoK Neurochemistry 008
Which among the following statements favour the dopamine hypothesis of schizophrenia?

Select one:

- Amphetamines release Dopamine into the synapse and cause positive schizophrenic symptoms
- Disulfiram stimulates Dopamine metabolism and may exacerbate schizophrenia
- Postal Emission Tomography (PET) shows decreased Dopamine receptor binding in drug naïve patients.
- Phencyclidine (PCP) causes negative schizophrenic symptoms
- Both alpha and beta isomers of Flupenthixol are effective anti psychotic drugs

Your answer is incorrect.

Only the alpha isomer of Flupentixol is a Dopamine antagonist and an effective anti psychotic drug. Phencyclidine causes both positive and negative symptoms of schizophrenia but via antagonisms at the NMDA receptor. Disulfiram actually inhibits Dopamine metabolism and may exacerbate schizophrenia. PET studies show increased Dopamine receptor binding in drug-naive patients. All these findings support the Dopamine hypothesis of schizophrenia.

The correct answer is: Amphetamines release Dopamine into the synapse and cause positive schizophrenic symptoms

Question 63

Not answered

Marked out of 1.00

Flag question

MoK Neurochemistry 010
Release of transmitters synthesized by neurons in the central nervous system is directly regulated by the concentration of which of the following electrolytes?

Select one:

- Calcium influx
- Chloride influx
- Potassium influx
- Magnesium influx
- Sodium influx

Your answer is incorrect.

Action potentials at the presynaptic axon terminals open up calcium channels, permitting calcium influx. This leads to movement of synaptic vesicles and subsequent release of stored neurotransmitters.

The correct answer is: Calcium influx

Question 64

Not answered

Marked out of 1.00

Flag question

MoK Neuropathology 004
A 75-year-old gentleman with memory problems, disinhibition and lack of insight. What microscopic findings will be expected in this case?

Select one:

- Spongiform encephalopathy
- Abnormally phosphorlyated tau protein
- Intraneuronal inclusions
- Beta amyloid peptides
- Cytoplasmic inclusions

Your answer is incorrect.

This man is exhibiting symptoms of fronto-temporal dementia. The syndrome of FTD has been associated with two types of pathological change, which in individual cases cannot be predicted on clinical grounds. Tau positive inclusions account for the majority of cases, and most familial (but not sporadic) cases with tau pathology carry mutations in the microtubule-associated tau protein on chromosome 17. The other main pathological finding in FTD is so-called ubiquitin-only pathology with cytoplasmic and nuclear inclusions (i.e. negative for tau or alpha-synuclein), suggesting a link with Amyotrophic lateral sclerosis.

The correct answer is: Cytoplasmic inclusions

Question 65

Not answered

Marked out of 1.00

Flag question

MoK Neuropathology 005
A 37-year-old male presents with episodes of waxing and waning neurological deficits. The most likely pathological change in brain is

Select one:

- Gliosis
- Demyelination
- Lewy bodies

- Neuritic plaques
- Amyloid plaques

Your answer is incorrect.

This patient has a diagnosis of Multiple Sclerosis. This is caused by the gradual demyelination that leads to delayed and inhibited signal conduction.

The correct answer is: Demyelination

Question 66
Not answered
Marked out of 1.00
Flag question

MoK Neuropathology 006
A newborn suffers from seizures and hydrocephalus; on examination midline facial anomaly and spasticity are evident. Further investigations reveal agenesis of corpus callosum. Which of the following underlying pathology is most likely?

Select one:
- Tissue loss secondary to obstetric trauma
- Disorder of neuronal migration
- Congential transmission of infection
- Excessive synaptic pruning
- Neurodegeneration

Your answer is incorrect.

Agenesis of CC is a disorder of neuronal migration. http://www.ninds.nih.gov/disorders/neuronal_migration/neuronal_migration.htm

The correct answer is: Disorder of neuronal migration

Question 67
Not answered
Marked out of 1.00
Flag question

MoK Neurophysiology 004
Which of the following EEG wavelets have low amplitude and high frequency?

Select one:
- Sigma waves
- Theta waves
- Beta waves
- Spike waves
- Alpha waves

Your answer is incorrect.

Beta waves have high (more than 13Hz) frequency but low amplitude

The correct answer is: Beta waves

Question 68
Not answered
Marked out of 1.00
Flag question

MoK Neurophysiology 006
Which of the following is a gut hormone released from small intestine in response to fats and proteins and serves to regulate food intake?

Select one:
- Neurokinin
- Serotonin
- Capsaicin
- Substance P
- Cholecystokinin

Your answer is incorrect.

CCK inhibits food intake by stimulating the vagal nerve and inhibiting gastric emptying. Central CCK receptors in the brain can inhibit eating.

The correct answer is: Cholecystokinin

Question 69
Not answered
Marked out of 1.00
Flag question

MoK Neurophysiology 007
The suprachiasmatic nucleus plays an important role in which of the following functions?

Select one:
- Attention
- Long term memory
- Circadian rhythms
- Sexual behaviour
- Water and hunger

Your answer is incorrect.

The suprachiasmatic nucleus controls the biologic clock of internal circadian rhythms.

The correct answer is: Circadian rhythms

Question 70
Not answered
Marked out of 4.00
Flag question

MoK Pharmacology CXEMI004

Pharmacokinetics of antipsychotics
Identify the antipsychotic drug using descriptions below;

1. Half life more than 50 hours [ ]

2. Antipsychotics with a 'hit and run' profile (choose two drugs) [ ] [ ]

3. D2 antagonistic effect [ ]

Explanation: The half-life of antipsychotics varies greatly. Shorter half-life medications, such as quetiapine and ziprasidone, have 7-12 hours of half-life. Risperidone has a 3-hour half-life, but it is metabolized to 9-hydroxy risperidone (aka paliperidone), which has a 24-28-hour half-life while olanzapine has a 30-hour half-life. The longest half-life, about three days, is seen with aripiprazole.

Clozapine and quetiapine are unique atypical antipsychotic agents that 'hit' the D2 receptor (blockade) only for a short while to cause antipsychotic effects and then 'run' (dissociate) before they cause extrapyramidal side effects. As a result, they are recommended for use in Parkinson's disease with psychosis.

Amisulpride functions primarily as a D2 and D3 receptor antagonist. It may also have a unique 5HT-7 function, which is not fully understood.
Excerpts from Correll, CU. Psychiatry Weekly: Practical Dosing Strategies. Retrieved from http://www.psychweekly.com/aspx/article/articledetail.aspx?articleid=1107 (accessed April 6, 2015).

The correct answer is:
Half life more than 50 hours - Aripiprazole,

Antipsychotics with a 'hit and run' profile (choose two drugs) - Clozapine , Quetiapine,

D2 antagonistic effect - Amisulpride

Question 71
Not answered
Marked out of 3.00
Flag question

MoK Pharmacology EMI005
Uncommon adverse effects
For each of the adverse effects given below, identify the best drug associated with it;

Severe sweating [Choose...]

Torsades de pointes [Choose...]

Allergic dermatitis and photosensitivity [Choose...]

Your answer is incorrect.

SSRIs: Severe sweating especially nocturnally is seen in some patients; Terazosin is effective in counteracting sweating.

Droperidol: Neuroleptics can decrease cardiac contractility, increase circulating levels of catecholamines, and prolong atrial and ventricular conduction time. Low-potency drugs are more cardiotoxic than high-potency drugs. ECG shows QT and PR prolongation, blunting of the T waves, and ST depression. Thioridazine and droperidol, in particular, can cause torsade de pointes. Antipsychotic related sudden death may be due to cardiac arrhythmias or even seizures asphyxiation or malignant hyperthermia. Drugs causing QT prolongation are associated with more sudden deaths.

Chlorpromazine: Allergic dermatitis and photosensitivity can occur with low-potency agents such as Chlorpromazine. Long-term chlorpromazine use can cause blue-gray discoloration of skin areas exposed to sunlight. This is reversible.

The correct answer is: Severe sweating – SSRIs, Torsades de pointes – Droperidol, Allergic dermatitis and photosensitivity – Chlorpromazine

Question 72
Not answered
Marked out of 3.00
Flag question

MoK Pharmacology EMI006
Drug-related adverse effects
For the drugs given below, identify the most important adverse effect leading to highly restricted use and requirements for appropriate monitoring

Pimozide [Choose...]

Vigabatrin [Choose...]

Clozapine [Choose...]

Your answer is incorrect.

Pimozide is rarely used a first line treatment of schizophrenia. Its use has been associated with ventricular arrhythmias and sudden unexplained death. Sertindole is another antipsychotic drug associated with sudden cardiac deaths. Droperidol and Thioridazine are also associated with significant QT prolongation on ECG.

Vigabatrin is associated with visual field defects.

Clozapine induced agranulocytosis is 15 to 30 times higher than the risk associated with phenothiazines. At least 0.7% of Clozapine treated patients develop agranulocytosis, which is potentially fatal. The maximum risk is between 4 and 18 weeks, and after a year the risk is same as with phenothiazines. Weekly monitoring of the white cell count is required for 26 weeks in most countries, with the frequency decreasing to biweekly or monthly after that.

The correct answer is: Pimozide – QT prolongation on ECG, Vigabatrin – Visual field defects, Clozapine – Agranulocytosis

Question 73
Not answered
Marked out of 3.00
Flag question

Mok Pharmacology EMI007
Plasma steady state
For each of the following drugs, identify the time taken to achieve steady state plasma levels

Lithium [Choose...]

Clozapine [Choose...]

Sodium valproate [Choose...]

Your answer is incorrect.

Lithium takes up to five days to reach steady plasma level. Clozapine takes 2-3 days, the same time as sodium valproate. (Ref: Maudsley prescribing guidelines- Pg 3)

The correct answer is: Lithium – 1 week, Clozapine – 2-3 days, Sodium valproate – 2-3 days

Question 74

Not answered

Marked out of 3.00

Flag question

MoK Neurosciences EMI006

Types of Mutations

For the following mutations, select the most appropriate term

Creation of a new stop codon — Choose...

This mutation does not alter the amino acid product — Choose...

Substitution of a purine for a purine — Choose...

Your answer is incorrect.

A silent mutation causes no change in protein product - this is possible because a single amino acid is often coded by more than one triplet sequence. In a silent mutation, one triplet sequence is replaced by a different sequence but without changing amino acid product.

In missense mutation, the new mutant codon specifies a different amino acid with variable effects on final protein product. For example, haemophilia, sickle cell anaemia. In nonsense mutation, the new codon is UUA UGA or UAG, which signals 'stop' to the amino acid sequence resulting in a nonfunctional protein. In frame shift mutations, the deletion or insertion is not in multiples of three codons e.g. a segment of 5 bases deletion mutations.

Point mutations are usually substitutions where one base is replaced by another. It could be termed as transition if a purine is replaced by another purine or a pyrimidine replaced by another pyrimidine (e.g. A to G). It is called transversion if a purine is replaced by a pyrimidine or vice versa (e.g. A to T).

The correct answer is: Creation of a new stop codon – Nonsense, This mutation does not alter the amino acid product – Silent, Substitution of a purine for a purine – Transition

Question 75

Not answered

Marked out of 3.00

Flag question

MoK Neurosciences EMI007

Chromosomal loci

Select the most appropriate chromosomal locus for each of the scenario given below

A 7-year-old girl has a large mandible and an open-mouthed expression which reveals the tongue. She has severe learning difficulties, with poor communication skills and little or absent speech. She has an unusual manner of laughing when stimulated. — Choose...

A 3-year-old girl repeatedly presents with respiratory infections. On examination, her larynx appears to be maldeveloped, and she has an unusually small head with a round face and a small chin with folds of skin over her eyes. The mother recalls a distinctive crying sound that became less noticeable after age 2. — Choose...

A 4-year-old girl is brought to the A&E with severe breathing problems. On examination, there is excessive drooling, with reduced muscle tone in her limbs. Developmental history is normal up to the age of 18 months apart from repeated problems related to gastroesophageal reflux. Growth charts reveal a slowing of head growth beginning at approximately 5 - 6 months of age. — Choose...

Your answer is incorrect.

Case 1 describes Angelman syndrome. A small deletion of part of chromosome 15 inherited from mother accounts for about 70 percent of those with Angelman syndrome. A similar genetic fault to the chromosome inherited from the father is found in another genetic condition, Prader-Willi syndrome.

Case 2 exemplifies Cri-du-chat syndrome. Cri-du-chat is caused by a variable length microdeletion on the short arm of chromosome 5. One of the deleted genes codes for TERT (telomerase reverse transcriptase) enzyme.

Case 3 is an example of Rett syndrome associted with a defect in the methyl-CpG-binding protein 2 (MeCP2) genes on the X chromosome. As females have two X chromosomes, even when one has this significant defect, the other X chromosome is normal enough for the child to survive. But most of the male fetuses die and so Rett's syndrome is almost exclusively restricted to female children.
Rett syndrome: MedlinePlus Medical Encyclopedia, http://www.nlm.nih.gov/medlineplus/ency/article/001536.htm (accessed April 5, 2015).

The correct answer is: A 7-year-old girl has a large mandible and an open-mouthed expression which reveals the tongue. She has severe learning difficulties, with poor communication skills and little or absent speech. She has an unusual manner of laughing when stimulated. – Chromosome 15, A 3-year-old girl repeatedly presents with respiratory infections. On examination, her larynx appears to be maldeveloped, and she has an unusually small head with a round face and a small chin with folds of skin over her eyes. The mother recalls a distinctive crying sound that became less noticeable after age 2. – Chromosome 5, A 4-year-old girl is brought to the A&E with severe breathing problems. On examination, there is excessive drooling, with reduced muscle tone in her limbs. Developmental history is normal up to the age of 18 months apart from repeated problems related to gastroesophageal reflux. Growth charts reveal a slowing of head growth beginning at approximately 5 - 6 months of age. – Chromosome X

Question 76

Not answered

Marked out of 6.00

Flag question

MoK Neurosciences CXEMI009

Neuropathology

For the following clinical conditions, identify the most common neuropathological changes in the brain.

1. A 70-year-old gentleman with history of parkinsonism, confusion and visual hallucinations (Choose TWO)

2. An 85-year-old lady with progressive memory loss for recent events and dysphasia, which has become more generalized. She also has a 20-years history of alcohol misuse (Choose TWO)

3. A 57-year-old gentleman presenting with altered behaviour, apathy, disinhibition and word finding difficulties. Mother and aunt died from a similar condition. (Choose TWO)

Explanation

Case 1: This is a case of Lewy body dementia characterised by neuroleptic sensitivity and the presence of Lewy bodies on pathological examination. Lewy bodies are intraneuronal deposits.

Case 2 is an example of Alzheimer's disease characterized by amyloid plaques, neurofibrillary tangles and apo-E4 gene.

Case 3 describes frontotemporal dementia characterized by frontal and temporal lobe atrophy and mutations in the tau gene.

The correct answer is:
A 70-year-old gentleman with history of parkinsonism, confusion and visual hallucinations (Choose TWO) - Sensitivity to antipsychotics, Intraneuronal synuclein deposits in cortex and substantia nigra,

An 85-year-old lady with progressive memory loss for recent events and dysphasia, which has become more generalized. She also has a 20-years history of alcohol misuse (Choose TWO) - Presence of e-4 allele, Neurofibrillary tangles and amyloid plaques in neocortex,

A 57-year-old gentleman presenting with altered behaviour, apathy, disinhibition and word finding difficulties. His mother and aunt died from a similar condition. (Choose TWO) - Atrophy of frontal poles, Mutation in Tau gene

Question 77

Not answered

Marked out of 6.00

Flag question

MoK Neurosciences CXEMI010

Prefrontal syndromes

A 65-year-old man has presented to the clinic with features of frontal lobe deficits. Identify two common clinical features commonly seen in each type of pre-frontal syndrome

1. Medial prefrontal syndrome [ ] [ ]

2. Orbitofrontal syndrome [ ] [ ]

3. Dorsolateral prefrontal syndrome [ ] [ ]

Explanation

Medial (ventromedial) prefrontal syndrome, also called apathetic, or pseudo depressive type is featured by a paucity of spontaneous behaviours, sparse verbal output, reduced social knowledge and judgement of harmful intent.

The orbitofrontal syndrome is also called disinhibited or pseudopsychopathic type and is characterised by poor impulse control, explosive outburst, and inappropriate behavior.

Dorsolateral is called as dysexecutive or disorganized type is featured by cognitive dysfunction, diminished judgment, and planning and poor insight. The patients are also described to be concrete and inflexible in nature.

The correct answer is:
Medial prefrontal syndrome - Poverty of speech , Paucity of spontaneous behaviour,

Orbitofrontal syndrome - Poor impulse control, Explosive outbursts,

Dorsolateral prefrontal syndrome - Executive dysfunction, Diminished planning

Question 78

Not answered

Marked out of 9.00

Flag question

HiY Neurosciences CXEMI031

Chemistry of Neurotransmitters

Match the following chemical groups with the neurotransmitters in the list

1. Monoamines (3 answers) [ ] [ ] [ ]

2. Aminoacids (3 answers) [ ] [ ] [ ]

3. Peptides (3 answers) [ ] [ ] [ ]

Explanation:

Neurotransmitters include monoamines, aminoacids and peptides.

Monoamines include dopamine, serotonin, norepinephrine, epinephrine, acetyl choline and histamine.

Aminoacids include GABA, glycine and glutamate.

Peptides include endorphins, cholecystokinin, angiotensin-2, neurotensin and corticotrophin-releasing hormone.

The correct answer is:
Monoamines (3 answers) - Serotonin,Dopamine, Norepinephrine

Aminoacids (3 answers) - Glutamate, Glycine, GABA,

Peptides (3 answers) - Cholecystokinin, Endorphins, Angiotensin

Finish review